

Python Programming

We will be starting shortly

What song should I play today?



Quiz

Why are snakes hard to trick?

[answer given in lecture]



Coursework 1

Feedback returned



Coursework 2

Deadline on Monday 7pm



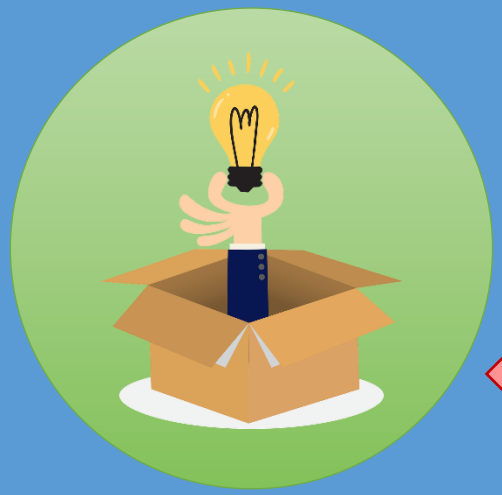
Object-Oriented Analysis and Design

(Software Engineering)

Understand & formulate problem

Implement algorithm







Test



Design algorithm to solve problem

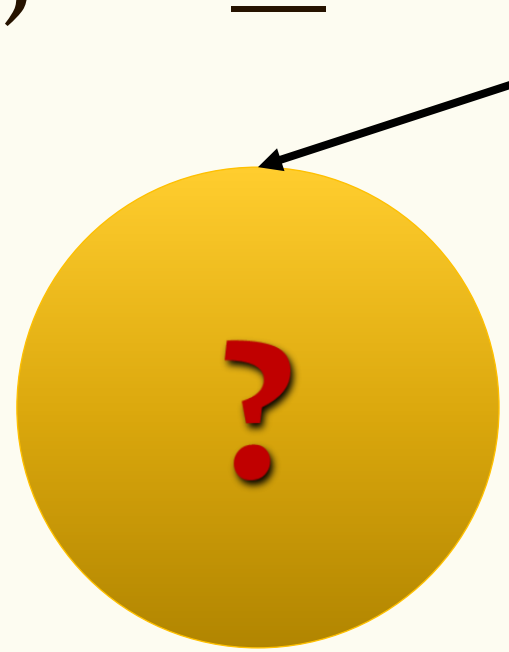


Object-Oriented Analysis and Design



Object-Oriented Analysis and Design

Object-Oriented Analysis and Design



Understand & formulate problem
(Identify objects and relations)

Object-Oriented Analysis and Design





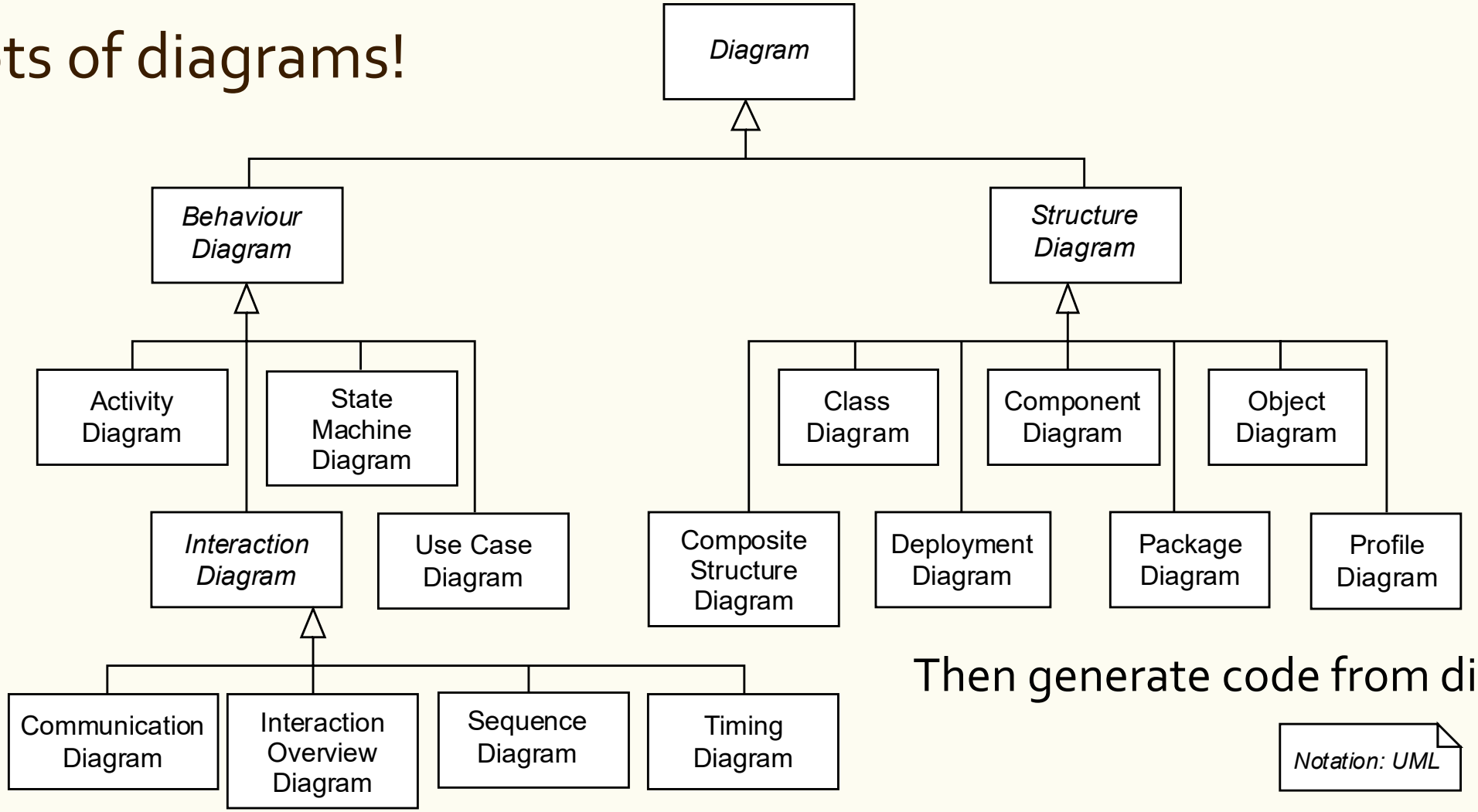
Understand & formulate problem
(Identify objects and relations)

Design ~~algorithm~~ to solve problem
classes and interactions



United Modelling Language (UML)

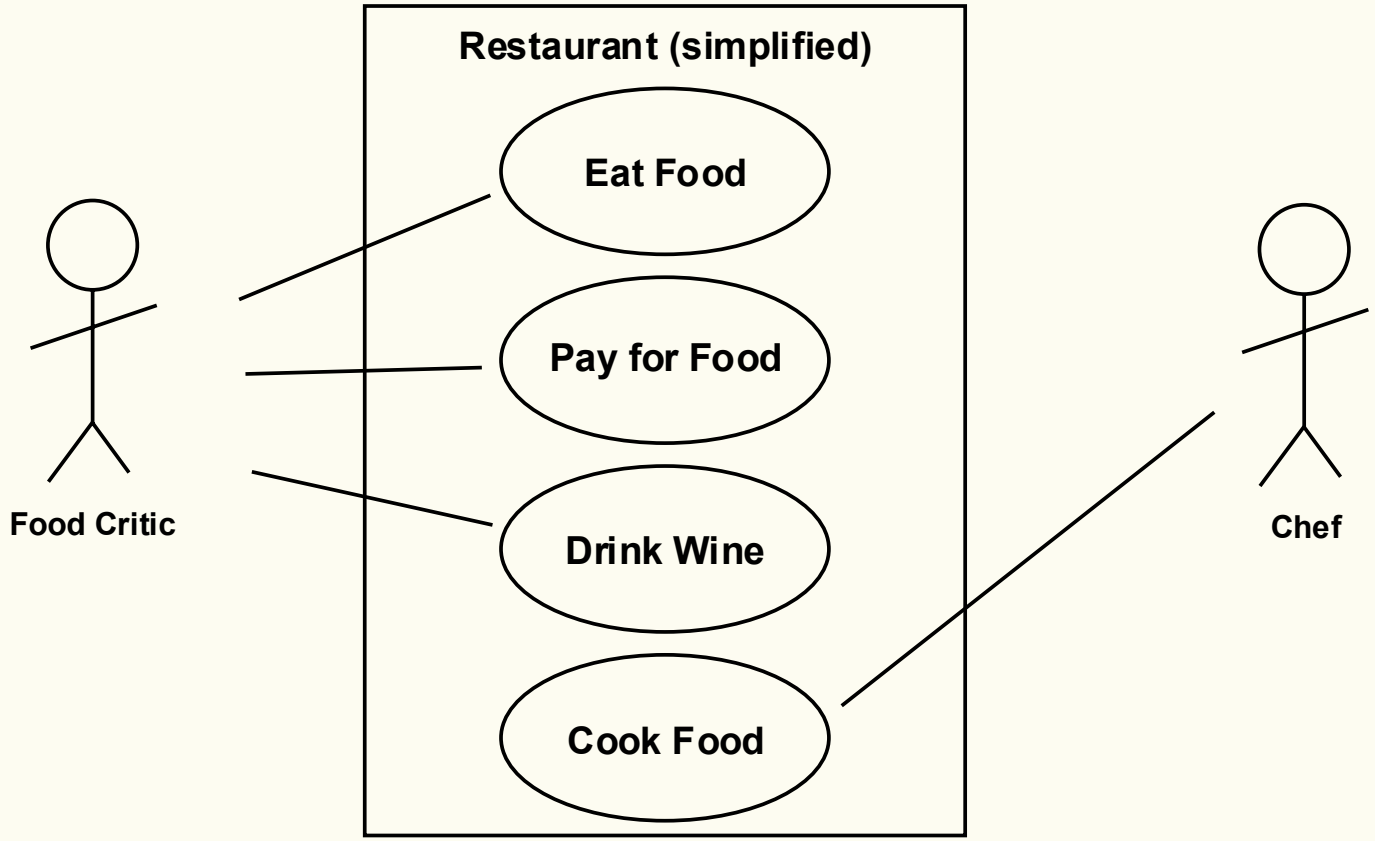
- Lots of diagrams!



Then generate code from diagrams

Notation: UML

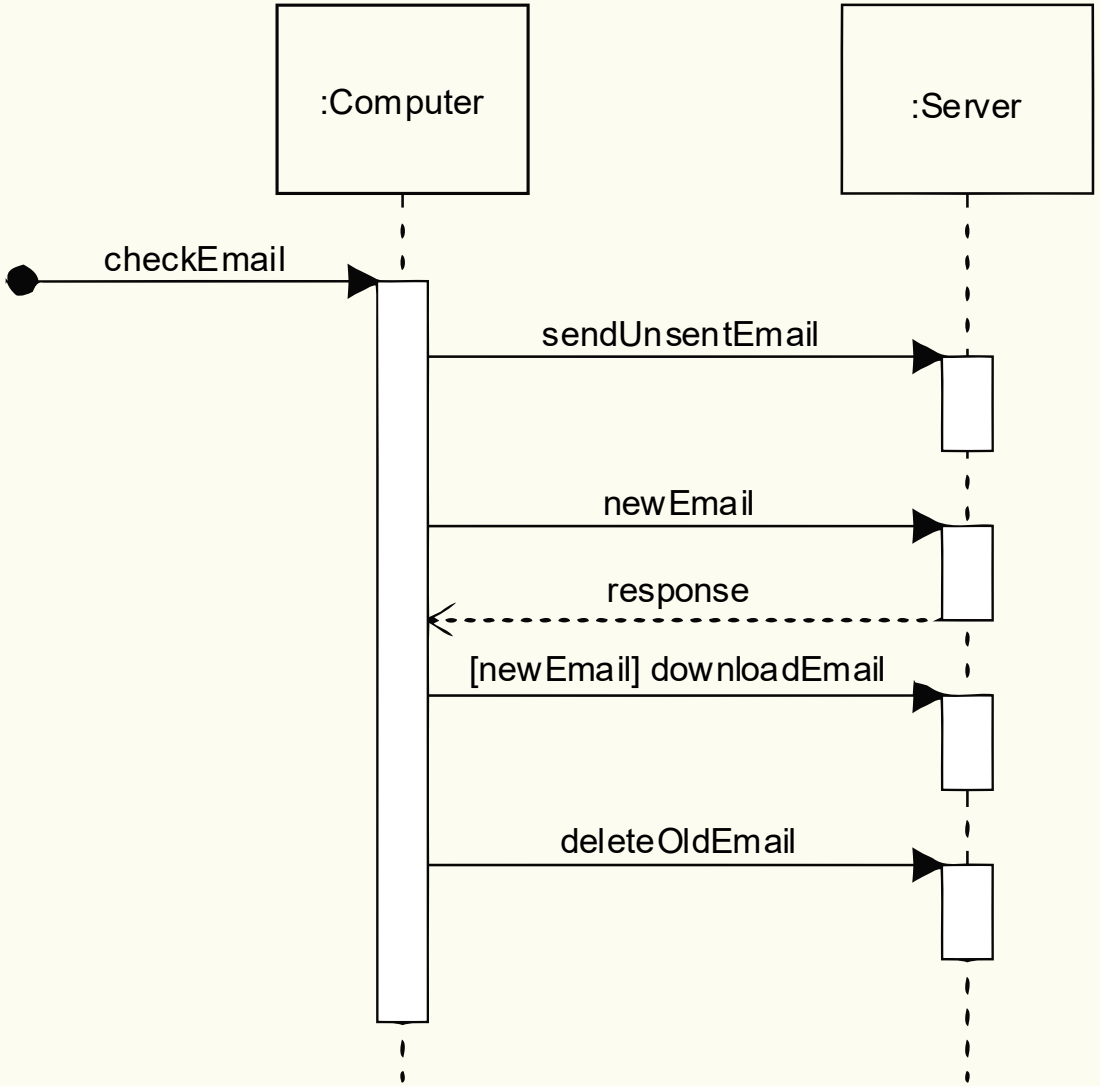
Use Case Diagram



Slashme, CC BY-SA 4.0, via Wikimedia Commons
https://commons.wikimedia.org/wiki/File:UML_Use_Case_diagram.svg



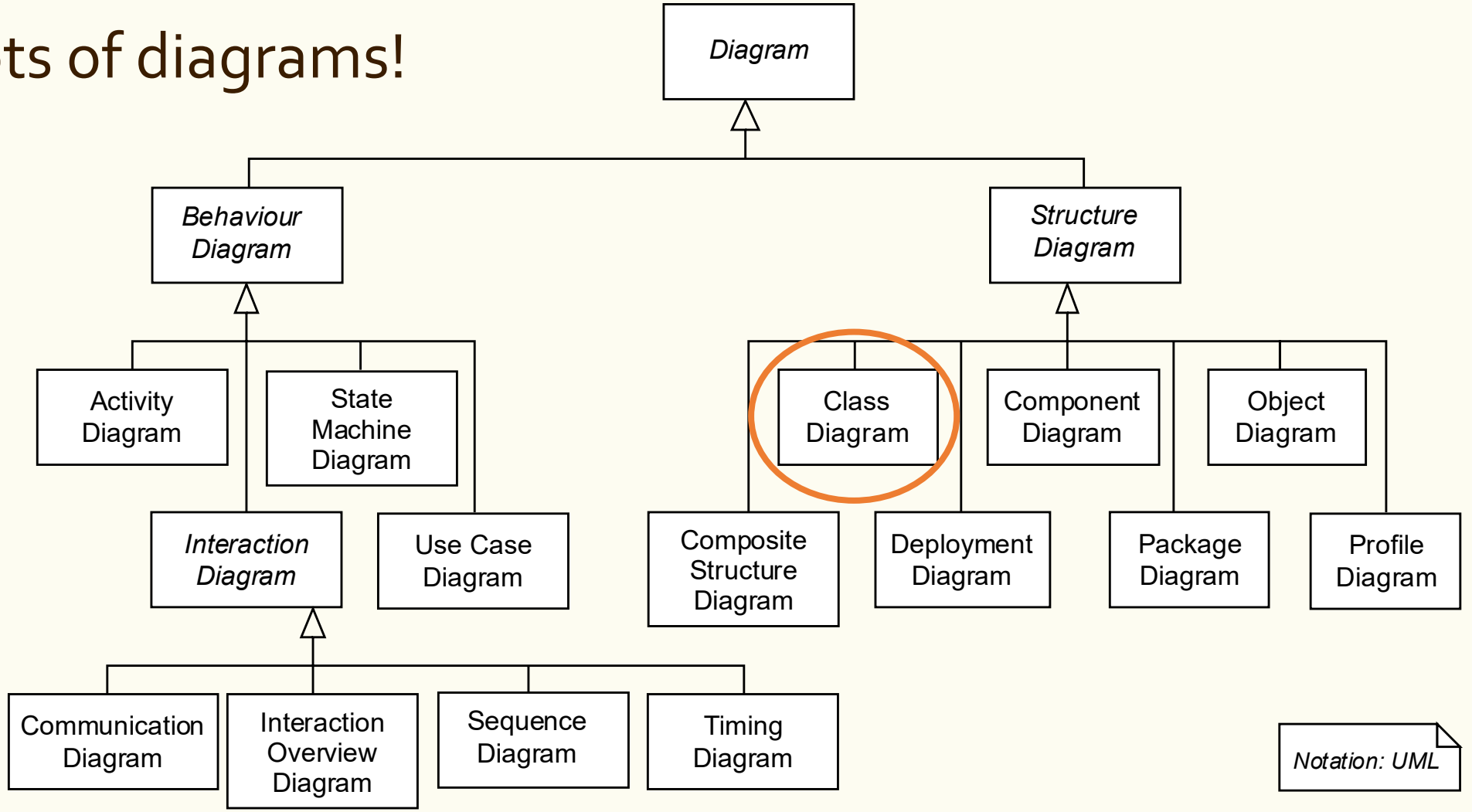
Sequence Diagram



Coupling_loss_graph.svg, CC BY-SA 3.0 via Wikimedia https://commons.wikimedia.org/wiki/File:CheckEmail.svg

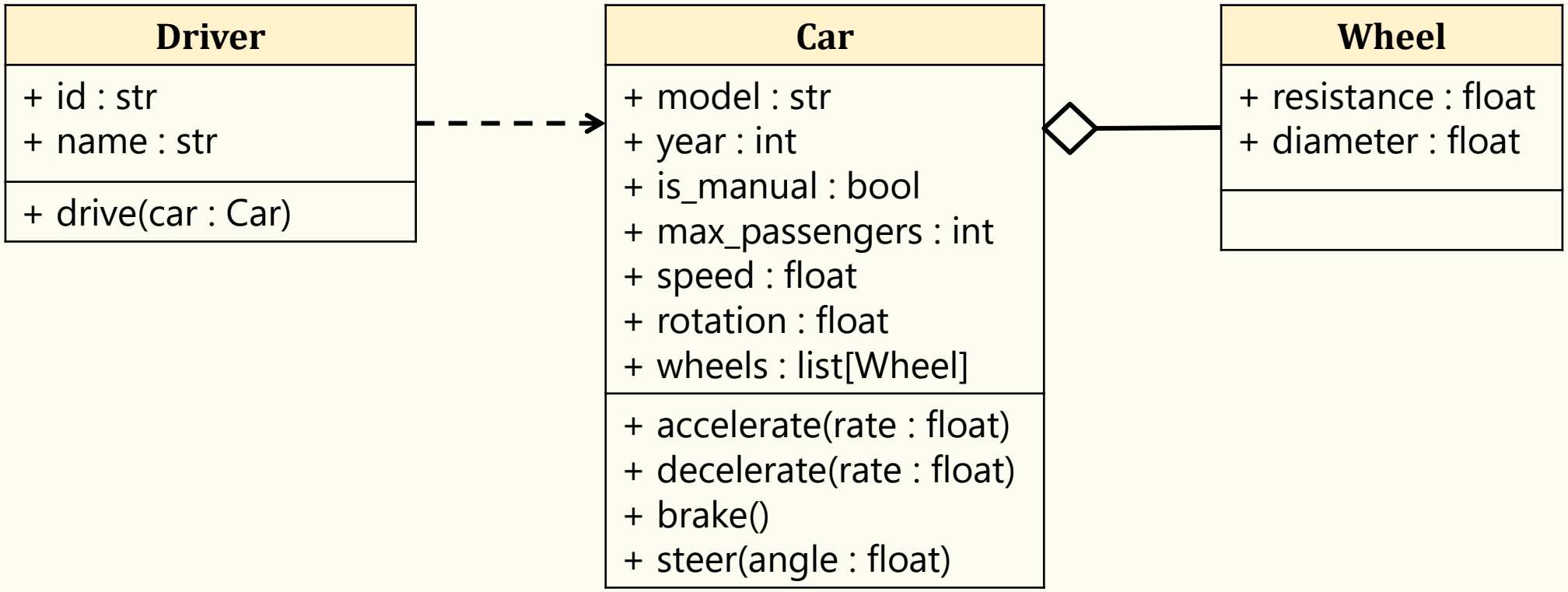
United Modelling Language (UML)

- Lots of diagrams!



United Modelling Language (UML)

- Focus on class diagrams





Group work

Perform OOAD given a scenario

(not much chance to do this)



- You are developing an online shopping app for a client.
- The client sells many different products. The products might be sold at a price that might change at any point (sales!)
- The client would like to be able to also show the original list price alongside the current sale price (you know, to entice customers)!

Any potential objects (and attributes)?



- Each customer will be assigned their own shopping cart. Customers should be able to retrieve the content of their shopping cart (the products and quantity) when they log in from any computer. They can add and delete products or update the quantities at any point until they check out.
- Upon checking out, the system will create a new order with an initial ‘confirmed’ state. For simplicity, you can assume that all payment has automatically been cleared when the customer checks out



- When the order has been prepared (the `ready for dispatch` state), it will generate a shipment with an initial estimated shipment date and an estimated arrival date. Once the order has been dispatched, the system should update the shipment date and estimated arrival date and set the order status to `dispatched’.



Group work

Produce a class diagram

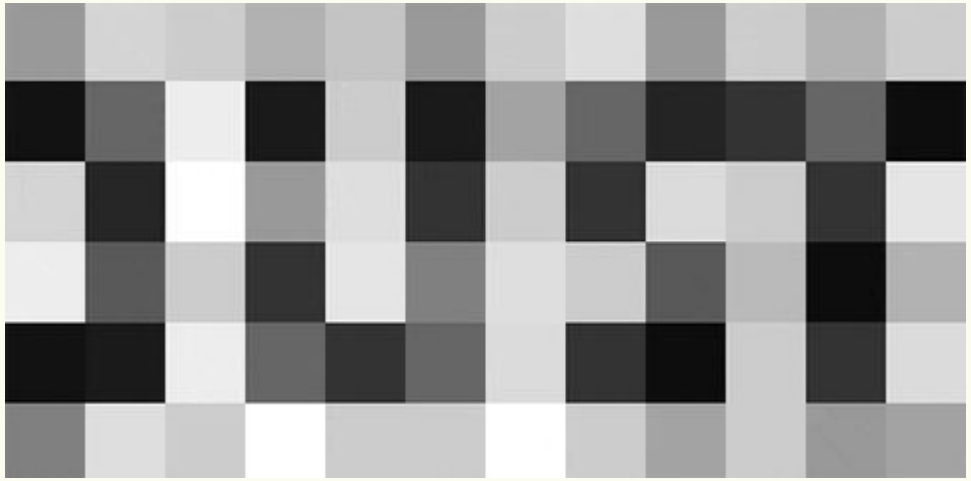
Just focus on ‘modelling’ the classes



- You are developing an online shopping app for a client.
- The client sells many different products. The products might be sold at a price that might change at any point (sales!) The client would like to be able to also show the original list price alongside the current sale price (you know, to entice customers)!
- Each customer will be assigned their own shopping cart. Customers should be able to retrieve the content of their shopping cart (the products and quantity) when they log in from any computer. They can add and delete products or update the quantities at any point until they check out.
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Solutions to be released after lecture







Summary

- Object-Oriented Analysis and Design
- United Modelling Language
  - Class diagrams
- Practise OOA&D on a concrete problem!



This and next week's schedule

CW2 due!

Mon 7pm GMT

Thu 4-6pm	Fri 11am-1pm
LAB 202/206	LAB 202/206

Tue 2-3pm	Tue 3-4pm	Thu 4-6pm	Fri 11am-1pm
LAB 219	LECTURE 144	LAB 202/206	LAB 202/206

Guest lecture on Deep Learning (by Konstantinos Mitsides)



Any (anonymous) feedback for us?

- https://forms.gle/NbaFwtvT2xhWbeSW7